



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8394**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	305	Seat No:	2022700063
Course Section:	1				

L41, 676, 986, L1505, 41482



Name of the candidate. **TRISHNA TIWARI**

Candidate's UID number:

2	0	2	2	7	0	0	0	6	3
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Examination class room number:

3	0	5
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Invigilator's signature with date:

 8/10

(To be filled in by the candidate)

EXAMINATION: **FY-BTECH**
(For example FY MCA/FY BTECH)

SEMESTER : **VII**

PROGRAM: **CSE-DS**

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: **Wednesday- 08/10/25**

COURSE NAME: **Deep Learning**

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q. 1]
a)

$$f(w) = \frac{1}{2} \|xw - y\|^2$$

Gradient update Equations are:-

$$1) w_{t+1} = w_t - \eta \nabla w_t$$

$$2) b_{t+1} = b_t - \eta \nabla b_t$$

Where w_t is weight at time t , η = Small learning rate.

∇w_t = change in weight update.

$$\nabla w_t = \frac{\partial L(w, b)}{\partial w}$$

Similarly, b_t = bias at t

$$\nabla b_t = \frac{\partial L(w, b)}{\partial b}$$

$$L(w, b) = \frac{1}{2} [f(x) - y]^2$$

$$\text{Where } f(x) = \frac{1}{1 + e^{-(wx+b)}}$$

Now for ∇w

$$\nabla w = \frac{\partial L(w, b)}{\partial w} = \frac{\partial}{\partial w} \left[\frac{1}{2} (f(x) - y)^2 \right]$$

$$= \frac{1}{2} \cdot 2 (f(x) - y) \cdot \frac{\partial}{\partial w} (f(x) - y)$$

$$= (f(x) - y) \cdot \frac{\partial}{\partial w} f(x)$$

$$= (f(x) - y) \cdot \frac{\partial}{\partial w} \left[\frac{1}{1 + e^{-(wx+b)}} \right]$$



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Q.1]

a)

$$= [f(x) - y] - \left(\frac{1}{1 + e^{(wx+b)}} \right)^2 \cdot \frac{\partial}{\partial w} (1 + e^{-wx+b})$$

$$= [f(x) - y] - \frac{1}{(1 + e^{-(wx+b)})^2} \cdot e^{-(wx+b)} \cdot \frac{\partial}{\partial w} (-wx+b)$$

$$= [f(x) - y] \cdot \frac{1}{[1 + e^{-(wx+b)}]^2} \cdot e^{-(wx+b)} \cdot (-x)$$

$$= [f(x) - y] \cdot \frac{e^{-(wx+b)}}{[1 + e^{-(wx+b)}]^2} \cdot \frac{1}{1 + e^{-(wx+b)}} \cdot x$$

$\underbrace{\frac{1}{1 + e^{-(wx+b)}}}_{f(x)}$

$$= [f(x) - y] \cdot [1 - f(x)] \cdot f(x) \cdot x$$

$$\therefore \nabla w = [f(x) - y] f(x) [1 - f(x)] x$$

$$\therefore \nabla b = [f(x) - y] f(x) [1 - f(x)]$$

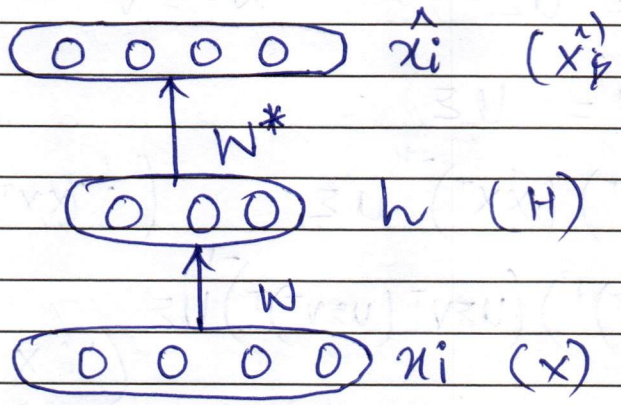


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Q.2]

b]

1] A linear undercomplete autoencoder is where $\text{dimension}(h) < \text{dimension}(x)$ where $x = \text{input layer}$ $h = \text{hidden layer}$. $\hat{x}_i = \text{output layer}$.



$$o/p = \min (x_{ij} - \hat{x}_{ij})^2$$

- 2] If $\hat{x}_i$ is almost similar to x_i , then h is a lossless layer and it captures all important features of the input.
- 3] which is similar to PCA where we reduce dimensions to get most important features

Setup:- linear encoder :- $H = XW$
linear decoder:- $\hat{X} = HW^*$

Normalize the inputs around the mean.

$$x_{ij} = \frac{1}{\sqrt{m}} (x_{ij} - \frac{1}{m} \sum x_{ij})$$

where x_{ij} = matrix elements
 m = Scaling factor.

let Original matrix = X & Reduced = x'

$$\therefore X = \frac{1}{\sqrt{m}} x'$$

Covariance Matrix = $x'^T x$

$$= \left(\frac{1}{\sqrt{m}} x' \right)^T \left(\frac{1}{\sqrt{m}} x \right)$$

$$= \frac{1}{m} (x')^T x \quad \parallel \rightarrow \textcircled{1}$$

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Q2]
b]

From S.V.D.
 $HW^* = U\Sigma V^T$

According to dimensions:

$$H = U\Sigma$$

$$W^* = V^T$$

$$H = U\Sigma$$

$$H = (XX^T)(XX^T)^{-1}U\Sigma \quad (\because (XX^T)(XX^T)^{-1} = I)$$

$$= (X(U\Sigma V^T)^T)(U\Sigma V^T(U\Sigma V^T)^T)^{-1}U\Sigma$$

($\because X = U\Sigma V^T$
- SVD)

$$= \cancel{X}(XV\Sigma^T U^T)(U\Sigma \underbrace{V^T V \Sigma^T U^T}_{I})^{-1}U\Sigma$$

$$= (XV\Sigma^T U^T)(U\Sigma \Sigma^T U^T)^{-1}U\Sigma \quad \{V^T V = I\}$$

$$= (XV\Sigma^T U^T)(U^T)^{-1}(\Sigma \Sigma^T)^{-1}(U)^{-1}U\Sigma$$

($\because (ABC)^{-1} = C^{-1}B^{-1}A^{-1}$)

$\because U$ matrix is Orthogonal
 $\therefore U^T = U^{-1}$

$$= (XV\Sigma^T U^T)(U^{-1})^{-1}(\Sigma \Sigma^T)^{-1}U^T U \Sigma$$

$$= XV \underbrace{\Sigma^T U^T U}_{I} (\Sigma \Sigma^T)^{-1} \underbrace{U^T U}_{I} \Sigma$$

$$= XV \Sigma^T (\Sigma \Sigma^T)^{-1} \Sigma$$

$\{U^T U = I\}$

$$= XV \underbrace{\Sigma^T \Sigma^{-1}}_{I} \underbrace{\Sigma}_{I}$$

$\{\Sigma \Sigma^{-1} = I\}$

$$= XV$$

$$\therefore H = XV$$

$\therefore H$ is a linear combination of X & V
But $H = XW$
(linear encoder).



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Q.2] b]		$\therefore W = V$. Where, V is $X^T X$ matrix from SVD and P is principle component of Covariance matrix $= X^T X$. $\therefore V \approx P$ $\therefore W = P$. $\therefore$ Autoencoder learns the same subspace as PCA. 1] An overcomplete Autoencoder is where $\dim(h) > \dim(x)$. 2] In such case, the input is just copied to hidden layer, then to output. 3] There is no learning of model as such and this AE proves to be useless. 4] Thus we apply L_2 regularization in which we put a limitation on the weight (weight decay) and force the model to update weights after learning. $\therefore \text{Eqn} = (x_{ij} - \hat{x}_{ij})^2 + \lambda \ \theta\ ^2$. Where the learning parameter is λ and θ is the changing rate. 5] So, we slow down the weight updation by regularizing the weights.

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Q.3]

a)

1] RNN is a neural network which trains for sequential data. (Eg. Text, speech, sales, etc).

2] It has a memory which preserves the input from the previous hidden state and uses it in output of the next hidden state.

3] $f_i = [Wx_t + b_i]$ Input
 $f_m = [Wh_{t-1} + b_h]$ memory.

4] This RNN has limitations like vanishing and exploding gradient.

5] Vanishing gradient is when weights are too small (< 1) and upon repetitive multiplication, they shrink to zero.

6] Exploding gradient when weights are too large (> 1) and upon repetitive multiplication they increase and explode.

7] LSTM (Long-Short Term Memory) is an improved version of RNN which solves the problem of vanishing gradient by including cell state and gate.

8] Cell State - The long term memory of the sequence.

9] Hidden State - The short term memory - Gates :-

10] Forget Gate (f) - Decides what to forget from memory.

$$f_t = \sigma[W_h f + x_f] + b_f$$

11] Input Gate (i) - Decides what input to store.

$$i_t = \sigma[W_h i + x_i] + b_i$$

12] Candidate gate (c) - What can be stored as input

$$c_t = \sigma[W_h c + x_c] + b_c$$

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2]

a)

1] Convolution - When we size see small parts of image. Done by passing a filter over the image then sliding it after small regions are seen, and we extract features such as edges etc.

2] Pooling - It shrinks the size of image and finds only the most important data points.

Max Pooling - Finds the max value from each part of image.

Average Pooling.

3] Fully connected layer - After both are done, this does the task of making decision (Classification / Prediction).

$$\text{Output size} = \frac{n+2p-f}{s} + 1$$

Where n = input size p = padding f = filter.
 s = stride.

Output volume

$$\text{No. of parameters} = (f \times f \times c + 1) \times K$$

$$\text{Output volume} = \left(\frac{n+2p-f}{s} + 1 \right) \left(\frac{n+2p-f}{s} + 1 \right) \times K.$$

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Q.3]

b]

1] GRU (Gated Recurrent Unit) is an improved version of LSTM which is faster and occupies lesser memory.

2] It combines the forget and input gate of LSTM, into one update gate.

3] Also, instead of 2 hidden states, it just has one.

4] Update Gate (u) - Decides which past information to forget.

$$u_t = \sigma(W_h u + x_u)$$

5] Reset Gate (r) - Decides which information to keep.

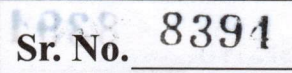
$$r_t = \sigma(W_h r + x_r)$$

6] Candidate - hidden state - Is the memory of the GRU and is a combination of candidate and cell state.

$$\# C_t = C_{t-1} x_t + f_t$$

7] GRU is for remembering information in which a long sequence of previous information.

8] It is faster as has lesser gates and saves up a lot of memory.

[illegible]



Sr. No. 8394

[illegible]

[illegible]